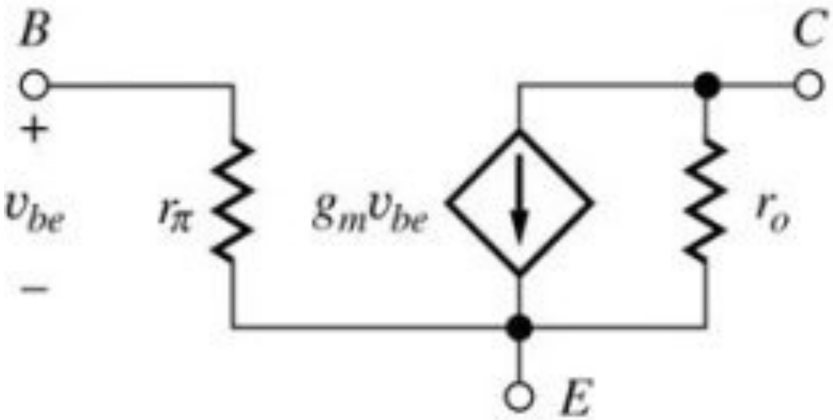
Chapter 6 – Single-Stage Amplifiers
Based on BJTs and MOSFETs

6.1 Single-Transistor Amplifiers

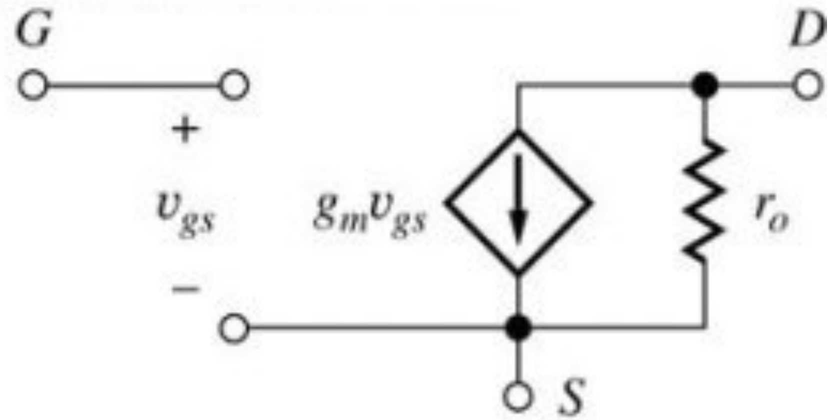
- Common-Emitter/Source Amplifiers
- Common-Base/Gate Amplifiers
- Common-Collector/Drain Amplifiers
- Single-Transistor Amplifier Comparison

Literature: Jaeger, Blalock, Chapter 14.1-14.7, page 857-925
Jaeger, Blalock, Chapter 15.1, page 968-991

Small Signal Models
– Y-Parameter/Admittance 2-Port Network –
– Example: C-E/C-S Amplifier –



BJT



MOSFET

Jaeger, Blalock, Fig. 14.6

- Looking at the small signal models of BJT and MOSFET, the MOSFET model is the same as the BJT model with $r_\pi = \infty$ (and $\beta_0 = \infty$ because $I_G = 0$)
- **Analyze BJT-based amplifier circuit first**; obtain results for FET-based amplifier circuit by setting $r_\pi = \infty$ and $\beta_0 = \infty$

6.1 Single-Transistor Amplifiers

6.1.1 Amplifier Classification
- Inverting Amplifiers
- Followers
- Non-Inverting Amplifiers

6.1.2 Common-Emitter/Source Amplifiers

6.1.3 Common-Base/Gate Amplifiers

6.1.4 Common-Collector/Drain Amplifiers

6.1.5 Single-Transistor Amplifier Comparison

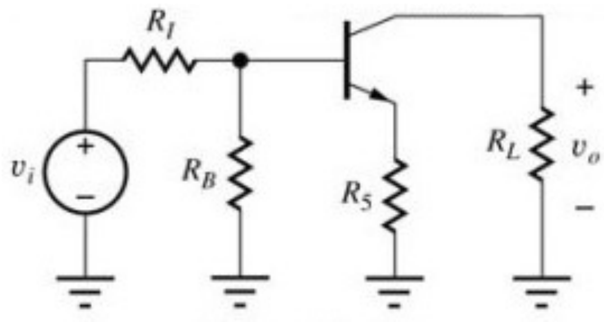
6.1.6 Coupling and Bypass Capacitors

Literature: Jaeger, Blalock, Chapter 14.1-14.7, page 857-925

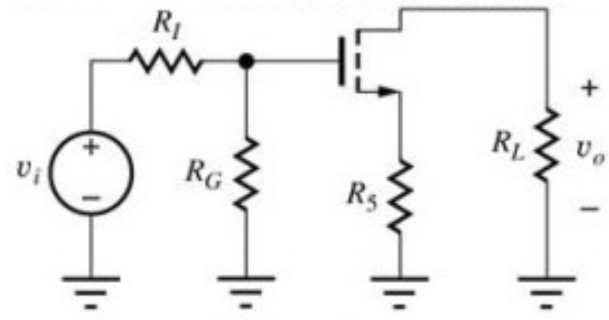
6.1.1 Amplifier Classification

Inverting Amplifiers

- Common-Emitter
- Common-Source



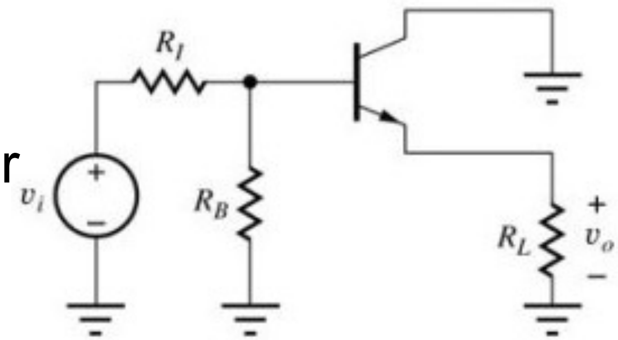
Common Emitter



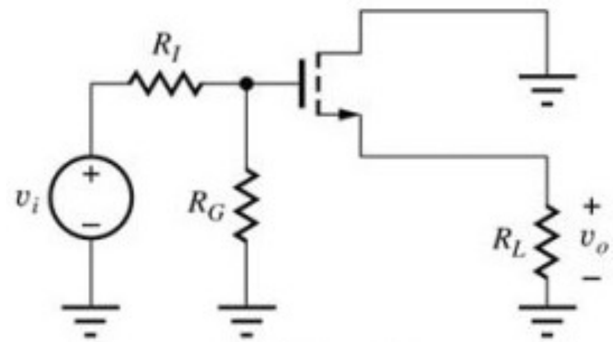
Common Source

Followers

- Common-Collector
- Common-Drain



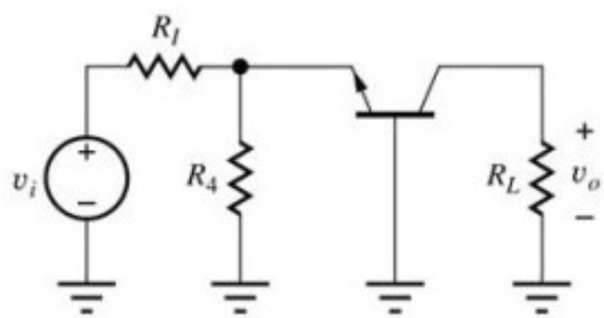
Common Collector



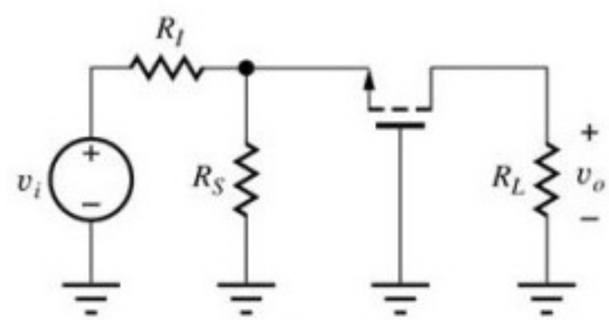
Common Drain

Non-Inverting Ampl.

- Common-Base
- Common-Gate



Common Base



Common Gate

Jaeger, Blalock, page 858

Signal Injection and Extraction

The BJT in Forward-Active Region

$$i_C = I_S\left[e^{qv_{BE}/kT}\right]$$

$$i_B = \frac{i_C}{\beta_F} = \frac{I_S}{\beta_F}\left[e^{qv_{BE}/kT}\right]$$

$$i_E = \frac{I_S}{\alpha_F}\left[e^{qv_{BE}/kT}\right]$$

$$v_{BE} = v_B - v_E$$

- Signal Injection: Base or Emitter (to vary either v_B or v_E)
- Signal Extraction: Emitter or Collector

The FET in Saturation Region

$$i_D = i_S = \frac{K_n}{2}\left[v_{GS} - V_T\right]^2$$

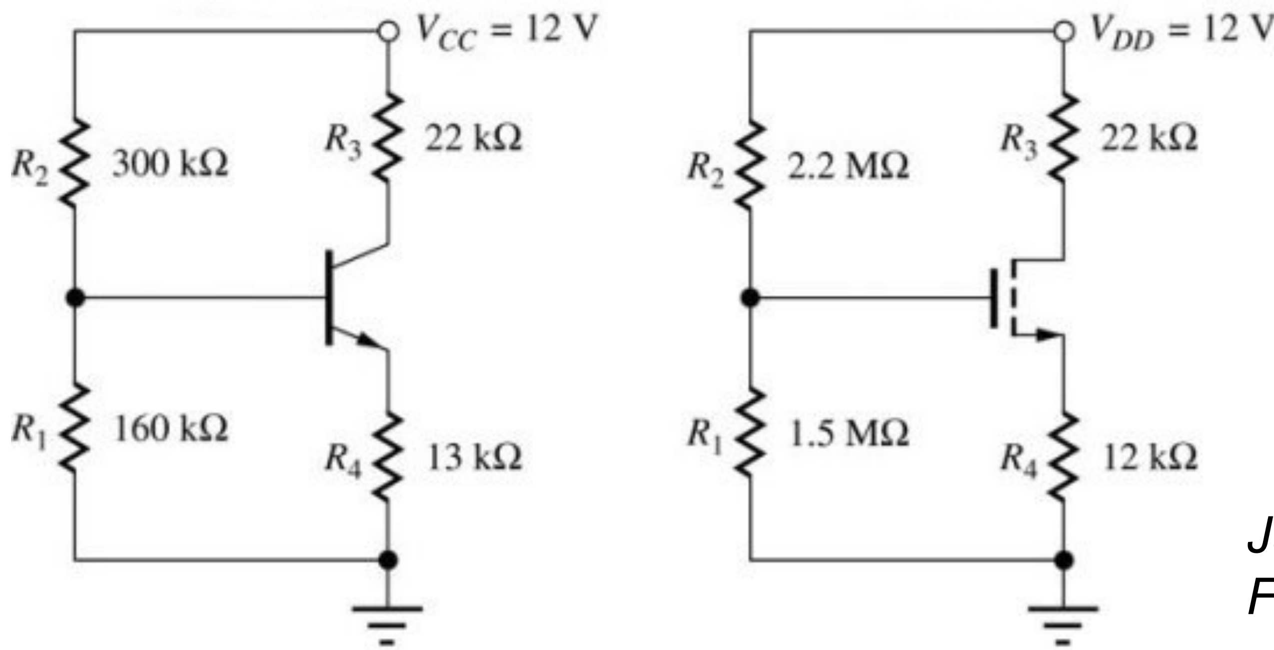
$$i_G = 0$$

$$v_{GS} = v_G - v_S$$

- Signal Injection: Gate or Source (to vary either v_G or v_S)
- Signal Extraction: Source or Drain

Four-Resistor Bias Circuit

- Goal: Achieve **similar Q-point for BJT- and MOSFET-based amplifier** circuits to allow direct performance comparison



Jaeger, Blalock, Fig. 14.1

- **MOSFET**: $K_n = 500\ \mu A/V^2$, $V_T = 1$ V, $\lambda = 0.02\ V^{-1}$ yield **(I_D, V_{DS}) = (241 μA, 3.81 V)**, $V_{GS} – V_T = 0.982$ V, $g_m = 0.49$ mS, $r_\pi = \infty$, and $r_o = 223$ kΩ
- **BJT**: $\beta_F = 100$, $V_A = 50$ V yield **(I_C, V_{CE}) = (245 μA, 3.64 V)**, $g_m = 9.80$ mS, $r_\pi = 10.2$ kΩ, and $r_o = 219$ kΩ

What do we compare?

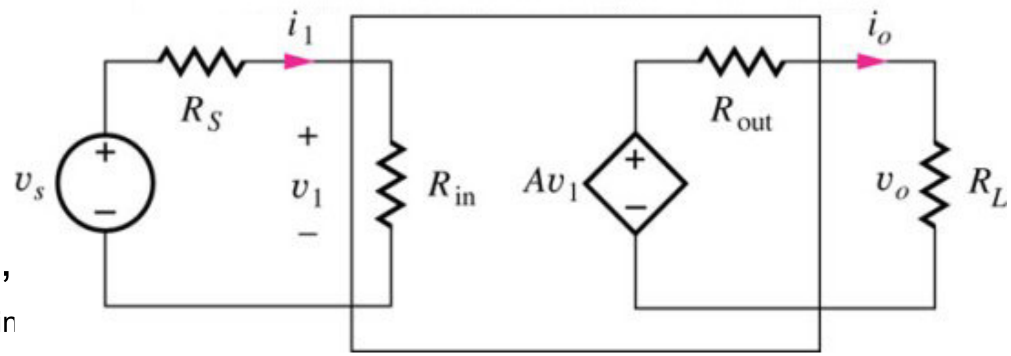
- Terminal Voltage Gain
- Signal Source Voltage Gain
- Current Gain
- Input Voltage Range
- Input and Output Resistance
- Coupling and Bypass Capacitor Design

Voltage Amplifier vs. Current Amplifier

Voltage Amplifier

$$\boxed{A_v = \frac{v_o}{v_s} = A\frac{R_{in}}{R_S + R_{in}}\frac{R_L}{R_{out} + R_L} \leq A}$$

- To achieve maximum voltage gain, the resistors should satisfy $R_S \ll R_{in}$ and $R_L \gg R_{out}$
- Ideal Voltage Amplifier: $R_{in} = \infty$, $R_{out} = 0$



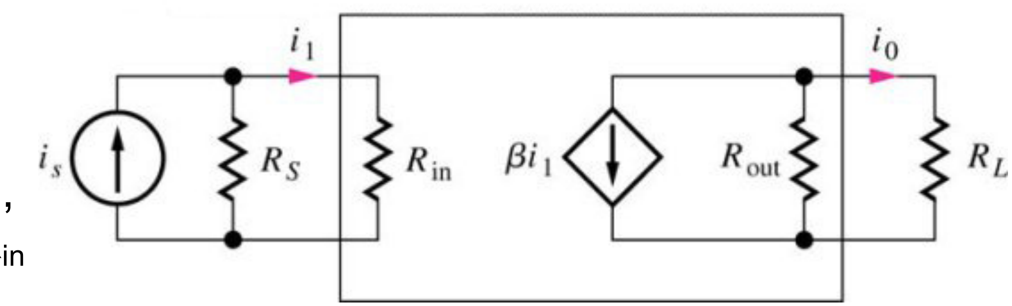
2-port g-parameter representation of voltage amplifier with source and load connected

Jaeger, Blalock, Fig. 10.9

Current Amplifier

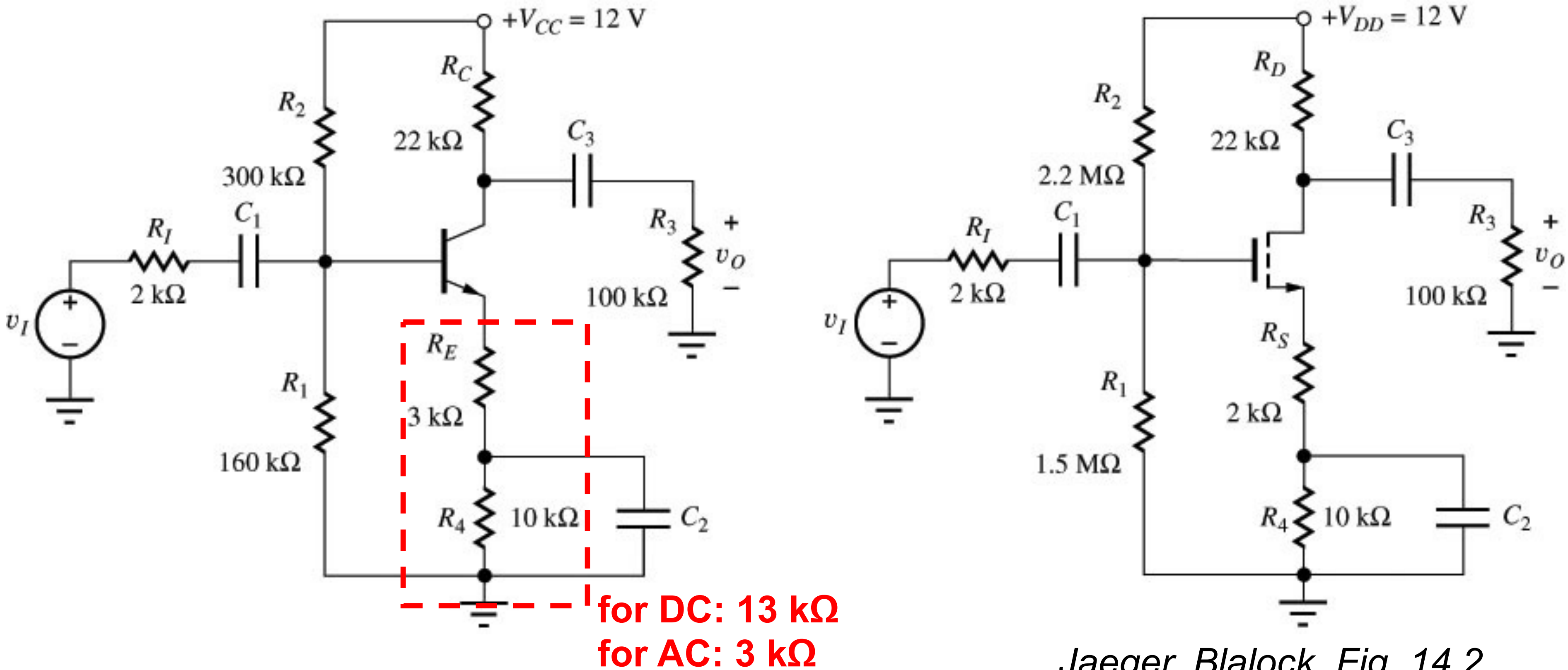
$$\boxed{A_i = \frac{i_o}{i_s} = \beta\frac{R_S}{R_S + R_{in}}\frac{R_{out}}{R_{out} + R_L} \leq \beta}$$

- To achieve maximum current gain, the resistors should satisfy $R_S \gg R_{in}$ and $R_L \ll R_{out}$
- Ideal Current Amplifier: $R_{in} = 0$, $R_{out} = \infty$



2-port h-parameter representation of voltage amplifier with source and load connected

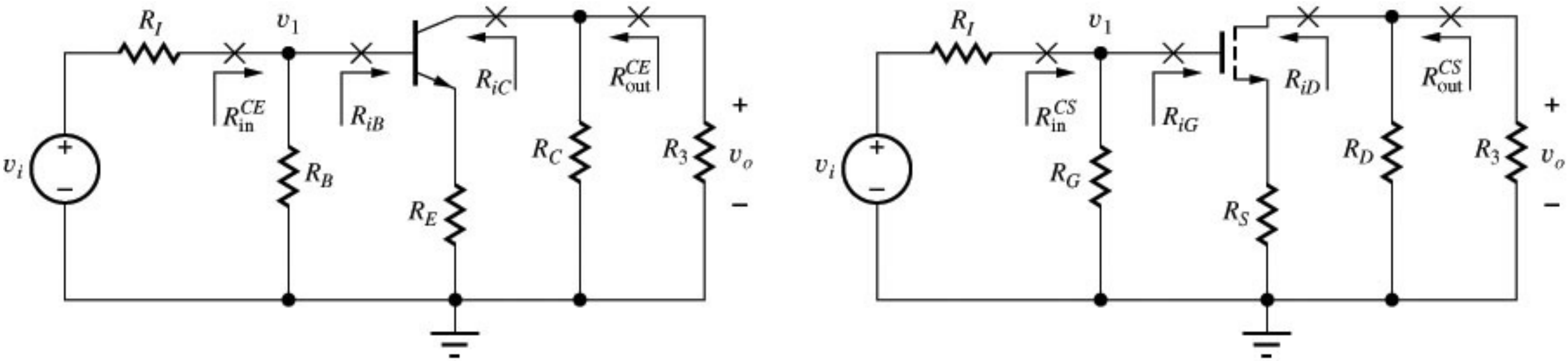
6.1.2 Common-Emitter/Source Amplifier



Jaeger, Blalock, Fig. 14.2

- Signal injection into base/gate, signal extraction from collector/drain
- Emitter/source as common terminal between input and output ports
- Only part of transistor's emitter/source resistor is bypassed by C_2 to gain flexibility in setting voltage gain, input/output resistance

Common-Emitter/Source Amplifier – Small-Signal Equivalent Circuit –



Jaeger, Blalock, Fig. 14.7

- BJT: $R_B = R_1||R_2 = 104\ k\Omega$, $R_L = R_C||R_3 = 18.0\ k\Omega$
  MOSFET: $R_G = R_1||R_2 = 892\ k\Omega$, $R_L = R_D||R_3 = 18.0\ k\Omega$
- **Terminal Voltage Gain A_{vt}:** voltage gain between base/gate and collector/drain terminals: $A_{vt} = v_o/v_b$ re. $A_{vt} = v_o/v_g$
- **Signal Source Voltage Gain A_v**: ratio output to input voltage $A_v = v_o/v_i$
- **Resistance R_{iB}/R_{iC}:** resistance R_{iB} looking into base terminal and resistance R_{iC} looking into collector terminal

Common-Emitter/Source Amplifier
– Terminal Voltage Gain –

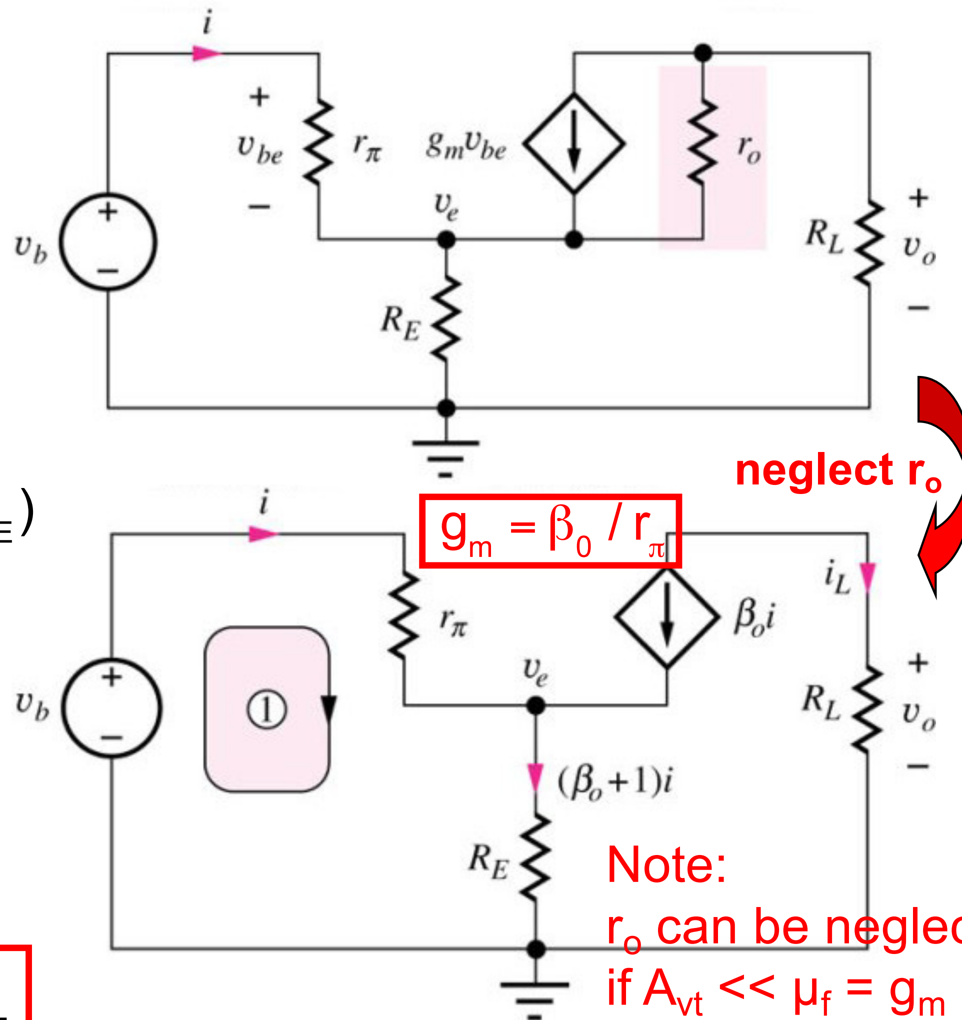
- r_o can be neglected during voltage gain analysis of resistively loaded amplifier (see Chapter 5.3)
- **Terminal Voltage Gain:**
  Output voltage: $v_o = -\beta_0 i R_L$
  KVL loop 1:

$$v_b = r_\pi i + (\beta_0 + 1) i R_E = i(r_\pi + (\beta_0 + 1) R_E)$$

$$A_{vt}^{CE} = \frac{v_o}{v_b} = \frac{-\beta_0 R_L}{r_\pi + (\beta_0 + 1) R_E}$$

$$= \frac{-\beta_0 R_L}{\frac{\beta_0}{g_m} + (\beta_0 + 1) R_E} \overset{\beta_0 \gg 1}{\cong} -\frac{g_m R_L}{1 + g_m R_E}$$

$$\boxed{A_{vt}^{CE} \cong -\frac{g_m R_L}{1 + g_m R_E} \qquad A_{vt}^{CS} \cong -\frac{g_m R_L}{1 + g_m R_S}}$$



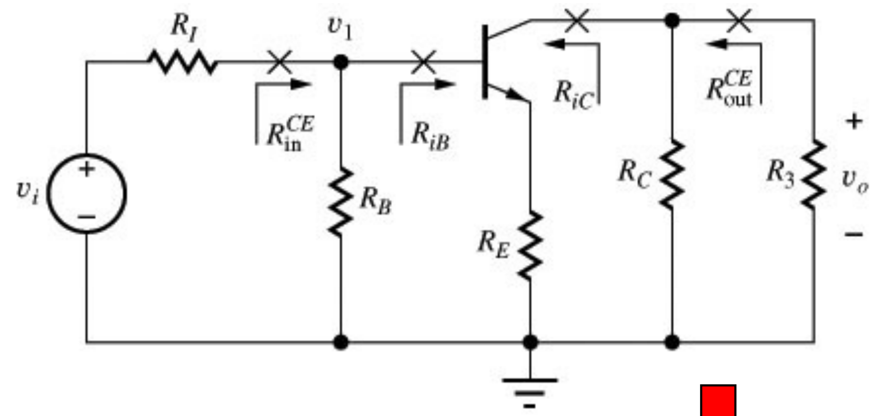
Note:
r_o can be neglected if $A_{vt} << \mu_f = g_m r_o$

Jaeger, Blalock, Fig. 14.8

Common-Emitter/Source Amplifier
– Input Resistance & Signal Source Voltage Gain –

- **Input Resistance**

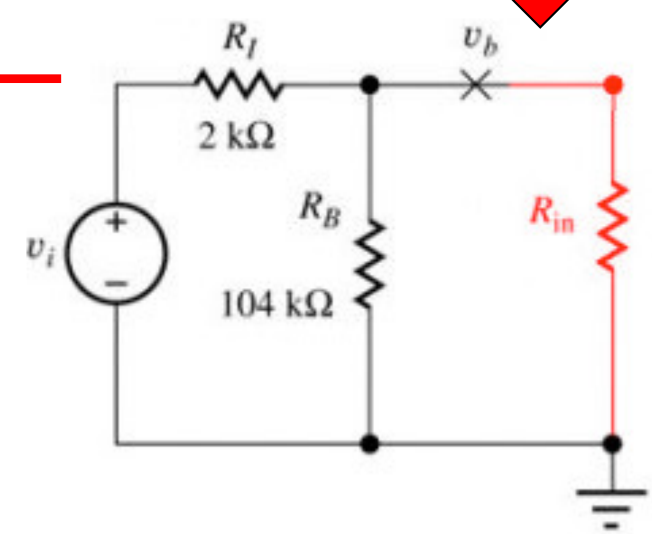
$$R_{iB} = \frac{v_b}{i} = r_\pi + (\beta_0 + 1)R_E \cong r_\pi(1 + g_m R_E)$$
$$R_{iG} = R_{iB}(r_\pi \to \infty) = \infty$$



- **Signal Source Voltage Gain:**

$$\underbrace{v_b = \frac{R_B \| R_{iB}}{R_I + R_B \| R_{iB}} v_i}$$

$$A_v^{CE} = \frac{v_o}{v_i} = \frac{v_o}{v_b}\frac{v_b}{v_i} = A_{vt}^{CE}\frac{v_b}{v_i} = A_{vt}^{CE}\frac{R_B \| R_{iB}}{R_I + R_B \| R_{iB}}$$
$$A_v^{CS} = \frac{v_o}{v_i} \overset{R_{iG}=\infty}{=} A_{vt}^{CS}\frac{R_G}{R_I + R_G}$$



Common-Emitter/Source Amplifier
– Voltage Gain Limits –

$$A_v^{CE} = \underbrace{\frac{R_B \,||\, R_{iB}}{R_I + R_B \,||\, R_{iB}}}_{\leq 1} A_{vt}^{CE} \qquad A_v^{CS} = \underbrace{\frac{R_G}{R_I + R_G}}_{\leq 1} A_{vt}^{CS}$$

➡ Overall amplifier gain $A_v \leq$ terminal voltage gain A_{vt}

$$A_{vt}^{CE} \cong -\frac{g_m R_L}{1 + g_m R_E} \qquad A_{vt}^{CS} \cong -\frac{g_m R_L}{1 + g_m R_S}$$

What are the (upper) limits for A_{vt}?

(a) $R_E = 0$, $R_S = 0$: $\quad A_{vt}^{CE} \cong -g_m R_L \qquad A_{vt}^{CS} \cong -g_m R_L$

(b) $g_m R_E \gg 1$, $g_m R_S \gg 1$: $\quad A_{vt}^{CE} \cong -\frac{R_L}{R_E} \qquad A_{vt}^{CS} \cong -\frac{R_L}{R_S}$

➡ $$\boxed{A_{vt}^{CE} \leq \min\left(-\frac{R_L}{R_E}, -g_m R_L\right) \qquad A_{vt}^{CS} \leq \min\left(-\frac{R_L}{R_S}, -g_m R_L\right)}$$

Common-Emitter/Source Amplifier
– Voltage Gain Limits (cont.) –

$$A_{vt}^{CE} \le \min\left(-\frac{R_L}{R_E}, -g_m R_L\right) \qquad A_{vt}^{CS} \le \min\left(-\frac{R_L}{R_S}, -g_m R_L\right)$$

- Emitter/source resistance reduces amplifier gain
- Why would a designer choose $R_E, R_S \neq 0$?
  - Voltage gain can be made substantially **independent of device parameter (e.g. g_m) variations**, and is given by resistance ratio R_L/R_E, R_L/R_S
  - Input/output resistance and input signal range can be improved (see below)

- What about the voltage gains of our example circuits?

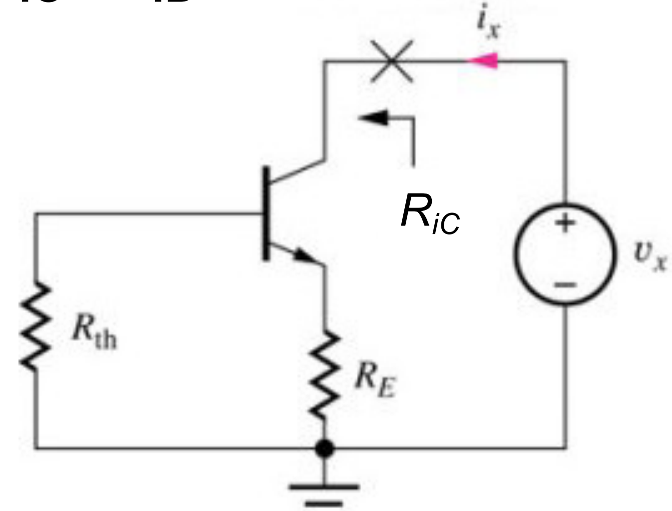
Common-Emitter/Source Amplifier
– Output Resistance R_{iC}, R_{iD}–

- Careful analysis of the circuit for the output resistance yields ($R_{th} = R_I \,||\, R_B$)

$$R_{iC} = r_o\left(1+\frac{\beta_0 R_E}{R_{th}+r_\pi+R_E}\right)+(R_{th}+r_\pi)\,||\,R_E$$

$$\cong r_o\left(1+\frac{\beta_0 R_E}{R_{th}+r_\pi+R_E}\right)$$

Amplification Factor:
$\mu_f = g_m r_o$

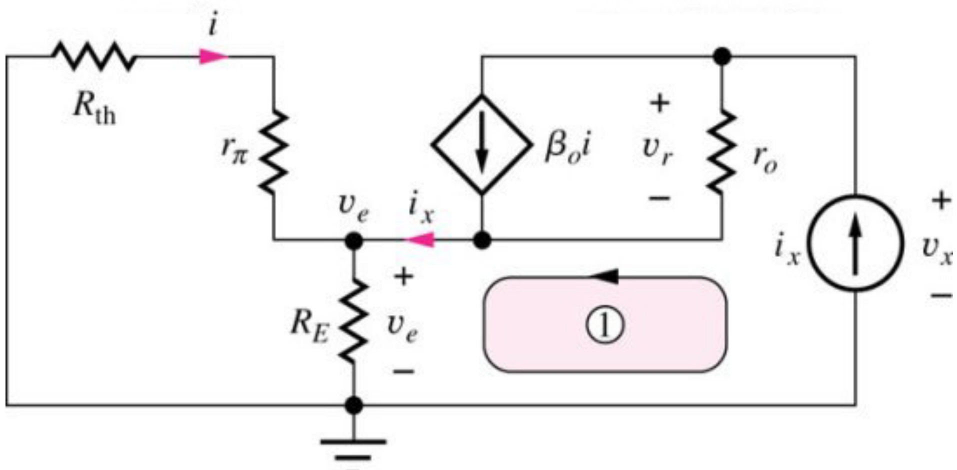


- Assuming $(r_\pi + R_E) \gg R_{th}$ and $\beta_0 = g_m r_\pi$, we find

$$R_{iC} = r_o\left(1+g_m\frac{r_\pi R_E}{r_\pi+R_E}\right)$$

$$= r_o\left(1+g_m(r_\pi \,||\, R_E)\right)$$

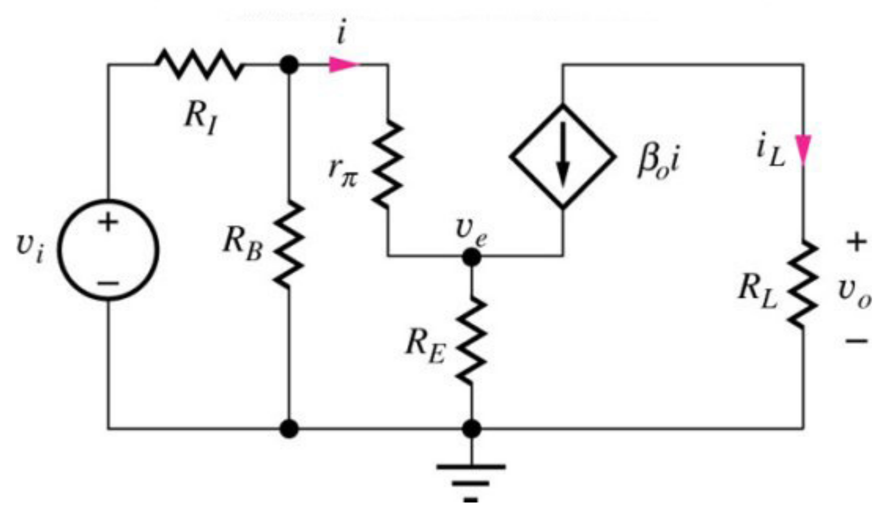
$$\overset{g_m(r_\pi||R_E)\gg 1}{\cong} r_o g_m(r_\pi \,||\, R_E) = \mu_f(r_\pi \,||\, R_E)$$



- **$R_{iC}(R_E=0) = r_o$, but can be designed to be » r_o for $g_m(R_E||r_\pi)$»1**

Common-Emitter/Source Amplifier
– Current Gain –

- Ratio of current i_L delivered to load resistor and current i supplied to base terminal:



$$A_{it} = \frac{i_L}{i}$$

- C-E amplifier:

$$i_L = -\beta_0 i \quad \Rightarrow \quad \boxed{A_{it}^{CE} = -\beta_0}$$

- C-S amplifier:

$$i = i_g = 0 \quad \Rightarrow \quad \boxed{A_{it}^{CS} = \infty}$$

Common-Emitter/Source Amplifier
– Performance Summary –

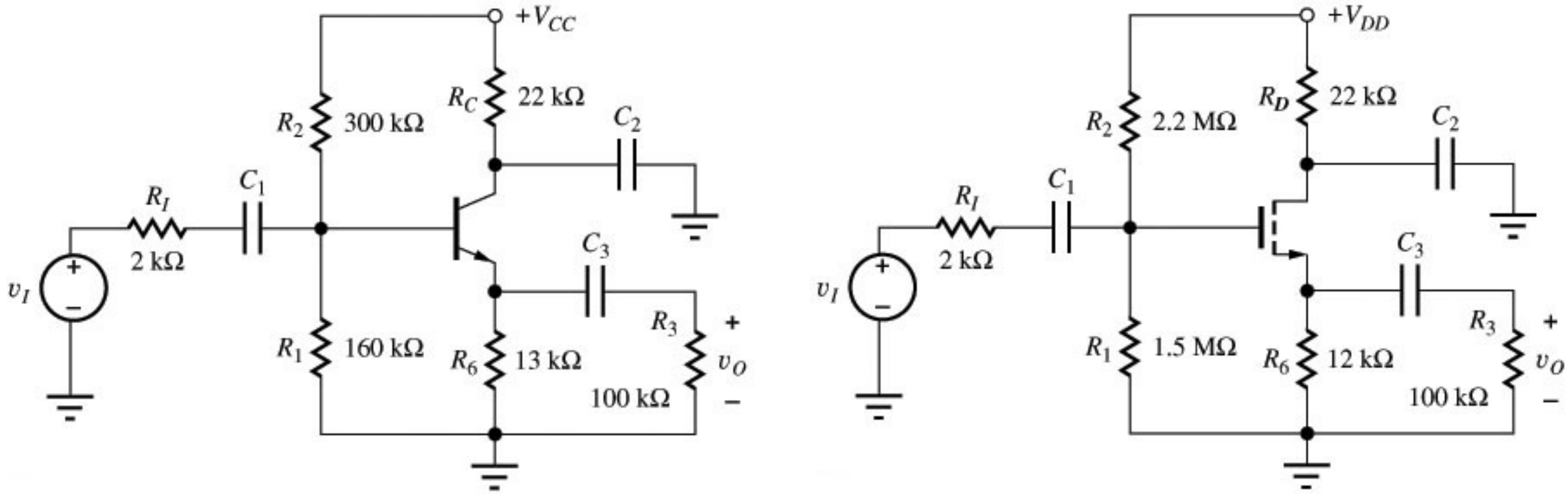
	C-E Amplifier	C-S Amplifier
Terminal Voltage Gain A_{vt}	$A_{vt}^{CE} \cong -\frac{g_m R_L}{1+g_m R_E}$	$A_{vt}^{CS} \cong -\frac{g_m R_L}{1+g_m R_S}$
Signal Source Voltage Gain A_v	$A_v^{CE} = -\frac{g_m R_L}{1+g_m R_E} \frac{R_B \Vert R_{iB}}{R_I + R_B \Vert R_{iB}}$	$A_v^{CS} = -\frac{g_m R_L}{1+g_m R_S} \frac{R_G}{R_I + R_G}$
Terminal Current Gain A_{it}	$A_{it} = -\beta_0$	$A_{it} = \infty$
Input Resistance R_{in}	$R_{iB} \cong r_\pi (1+g_m R_E)$	$R_{iG} = \infty$
Output Resistance R_{out}	$R_{iC} \cong r_o \left(1+g_m (r_\pi \Vert R_E)\right)$	$R_{iD} \cong r_o \left(1+g_m (r_\pi \Vert R_S)\right)$
Input Signal Range	$\lvert v_b \rvert \le 0.005\left(1+g_m R_E\right)$	$\lvert v_g \rvert \le 0.2\left(V_{GS} - V_T\right)\left(1+g_m R_S\right)$

Common-Emitter/Source Amplifier
– Performance Summary –

- C-E and C-S amplifiers feature high voltage gain and reasonably high input resistance
- Emitter/source resistor R_E, R_S reduces voltage gain but improves input/output resistance and usable input signal range
- Values for the amplifier circuits on page 8:

	C-E Amplifier	C-S Amplifier
$g_m R_{E/S}$	29.4	0.982
Voltage Gain	– 5.61	– 4.45
Input Resistance	310 kΩ	∞
Output Resistance	4.55 MΩ	442 kΩ
Input Signal Range	152 mV	389 mV
Current Gain	– 100	∞

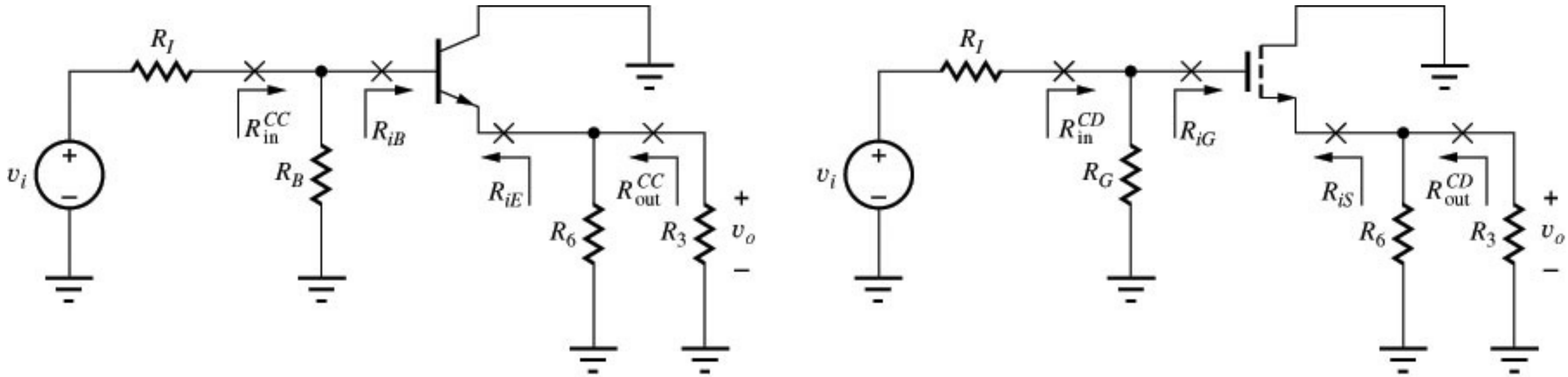
6.1.3 Common-Collector/Drain Amplifier



Jaeger, Blalock, Fig. 14.3

- Signal injection into base/gate, signal extraction from emitter/source
- Collector/drain as common terminal between input and output ports
- Collector/drain are bypassed directly to GND by C_2 to represent common AC GND for input and output port
- Note: In this circuit, the resistor R_C (or R_D) can be omitted without changing circuit characteristic!

Common-Collector/Drain Amplifier – Small-Signal Equivalent Circuit –

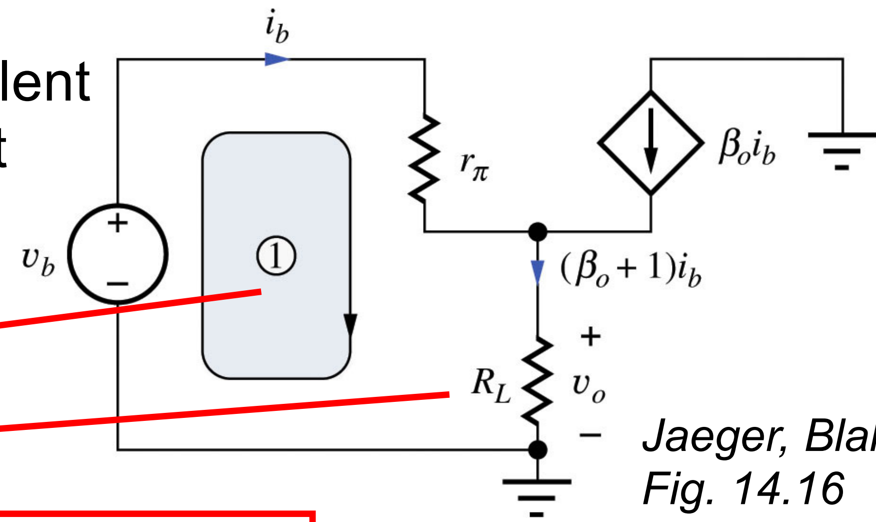


Jaeger, Blalock, Fig. 14.15

- BJT: $R_B = R_1||R_2 = 104\ k\Omega$, $R_L = R_6||R_3 = 11.5\ k\Omega$
  MOSFET: $R_G = R_1||R_2 = 892\ k\Omega$, $R_L = R_6||R_3 = 10.7\ k\Omega$
- C-C and C-D amplifiers have a gain that approaches 1; because the output voltage "follows" the input voltage (gain 1), C-C and C-D amplifiers are often called **emitter follower** and **source follower**, respectively

Common-Collector/Drain Amplifier – Terminal Voltage Gain –

- Analyze small signal equivalent circuit, neglecting the output resistance r_o



Jaeger, Blalock, Fig. 14.16

$$v_b = r_\pi i_b + (\beta_0 + 1) i_b R_L$$

$$v_o = (\beta_0 + 1) i_b R_L$$

$$\boxed{A_{vt}^{CC} = \frac{v_o}{v_b} = \frac{(\beta_0+1)R_L}{r_\pi + (\beta_0+1)R_L} \overset{\substack{\beta_0 \gg 1 \\ g_m = \beta_0/r_\pi}}{\cong} \frac{g_m R_L}{1+g_m R_L} \qquad A_{vt}^{CD} = \frac{g_m R_L}{1+g_m R_L}}$$

In most C-C and C-D designs, $g_m R_L \gg 1$, i.e.

$$\boxed{A_{vt}^{CC} \cong A_{vt}^{CD} \cong 1}$$

- Output resistance r_o can be included by replacing R_L with $R_L || r_o$

Common-Collector/Drain Amplifier
– Input Resistance and Signal Source Voltage Gain –

- **Input Resistance:** $v_b = r_\pi i_b + (\beta_0 + 1) i_b R_L$

$$\boxed{R_{iB} = \frac{v_b}{i_b} = r_\pi + (\beta_0 + 1) R_L \qquad R_{iG} \overset{r_\pi, \beta_0 \to \infty}{=} \infty}$$

- R_{in} of source follower is very large, R_{in} of emitter follower can be quite large

- **Signal Source Voltage Gain:**
  Calculated analogue to C-E voltage gain……

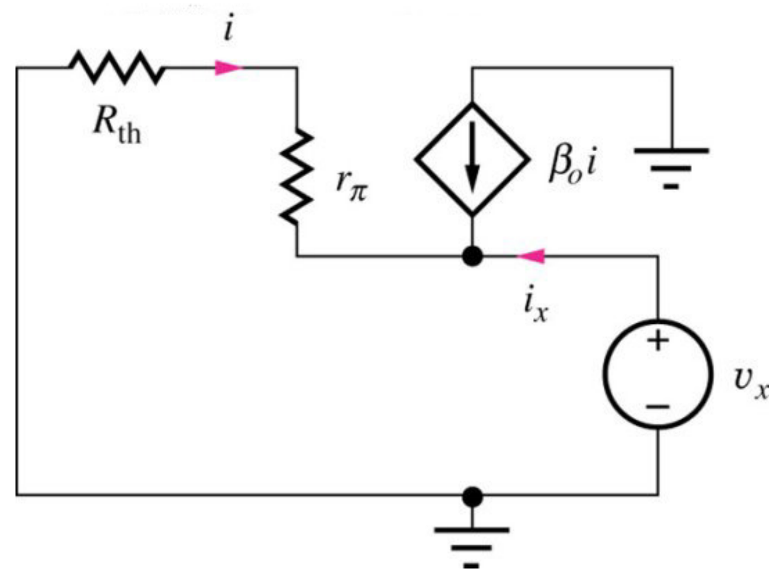
$$A_v^{CC} = \frac{v_o}{v_i} = \frac{v_o}{v_b}\frac{v_b}{v_i} = A_{vt}^{CC}\frac{v_b}{v_i} = A_{vt}^{CC}\frac{R_B \| R_{iB}}{R_I + R_B \| R_{iB}}$$

$$A_v^{CD} = \frac{v_o}{v_i} \overset{R_{iG}=\infty}{=} A_{vt}^{CD}\frac{R_G}{R_I + R_G}$$

Common-Collector/Drain Amplifier – Output Resistance R_{iE}, R_{iS} –

- Simple circuit analysis yields for the output resistance ($R_{th} = R_I \,||\, R_B$)

$$\boxed{R_{iE} = \frac{r_\pi + R_{th}}{\beta_0 + 1} = \frac{r_\pi}{\beta_0 + 1} + \frac{R_{th}}{\beta_0 + 1} \cong \frac{1}{g_m} + \frac{R_{th}}{\beta_0 + 1}}$$



Jaeger, Blalock, Fig. 14.17

- Because of infinite current gain of FET

$$\boxed{R_{iS} = \frac{1}{g_m}}$$

- Output resistance is dominated by reciprocal of transconductance; additional term for BJT is usually small (unless R_{th} is very large); the resulting output resistance is small

Common-Collector/Drain Amplifier
– Performance Summary –

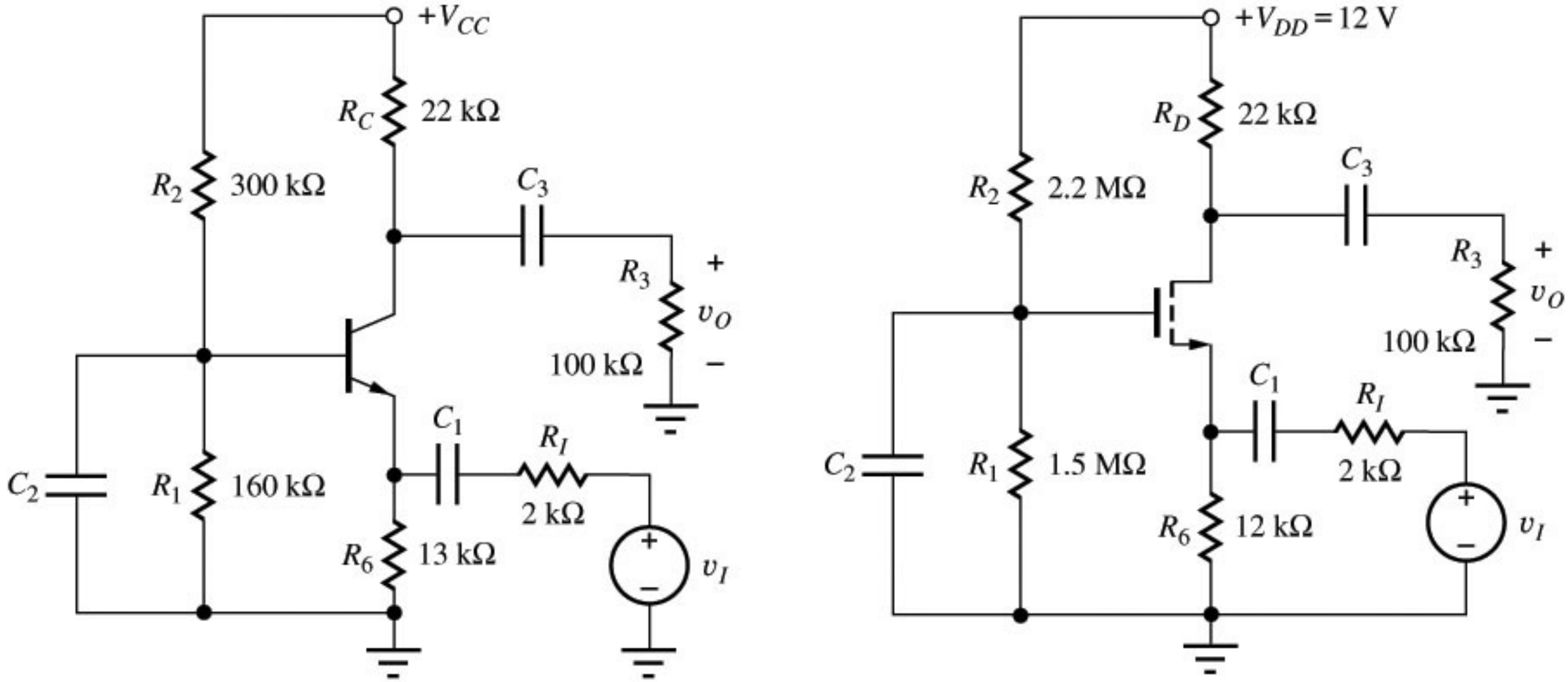
	C-C Amplifier	C-D Amplifier
Terminal Voltage Gain A_{vt}	$A_{vt}^{CC} \cong \frac{g_m R_L}{1+g_m R_L} \cong 1$	$A_{vt}^{CD} \cong \frac{g_m R_L}{1+g_m R_L} \cong 1$
Signal Source Voltage Gain A_v	$A_v^{CC} = \frac{g_m R_L}{1+g_m R_L} \frac{R_B \parallel R_{iB}}{R_I + R_B \parallel R_{iB}}$	$A_v^{CD} = \frac{g_m R_L}{1+g_m R_L} \frac{R_G}{R_I + R_G}$
Terminal Current Gain A_{it}	$A_{it} = \beta_0 + 1$	$A_{it} = \infty$
Input Resistance	$R_{iB} = r_\pi + (\beta_0 + 1) R_L$	$R_{iG} = \infty$
Output Resistance	$R_{iE} \cong \frac{\alpha_0}{g_m} + \frac{R_{th}}{\beta_0 + 1} \cong \frac{1}{g_m}$	$R_{iS} = \frac{1}{g_m}$
Input Signal Range	$\lvert v_b \rvert \le 0.005\left(1+g_m R_L\right)$	$\lvert v_g \rvert \le 0.2\left(V_{GS} - V_T\right)\left(1+g_m R_L\right)$

Common-Collector/Drain Amplifier
– Performance Summary –

- C-C and C-D amplifiers are used as voltage buffers with a gain of ≈ 1, very high input resistance and relatively small output resistance
- For ideal "follower" characteristic $g_m R_L \gg 1$
- Values for the amplifier circuits on page 19:

	C-C Amplifier	C-D Amplifier
$g_m R_L$	113	5.25
Voltage Gain	0.956	0.838
Input Resistance	1.16 MΩ	∞
Output Resistance	121 Ω	2.04 kΩ
Input Signal Range	0.575 V	1.23 V
Current Gain	101	∞

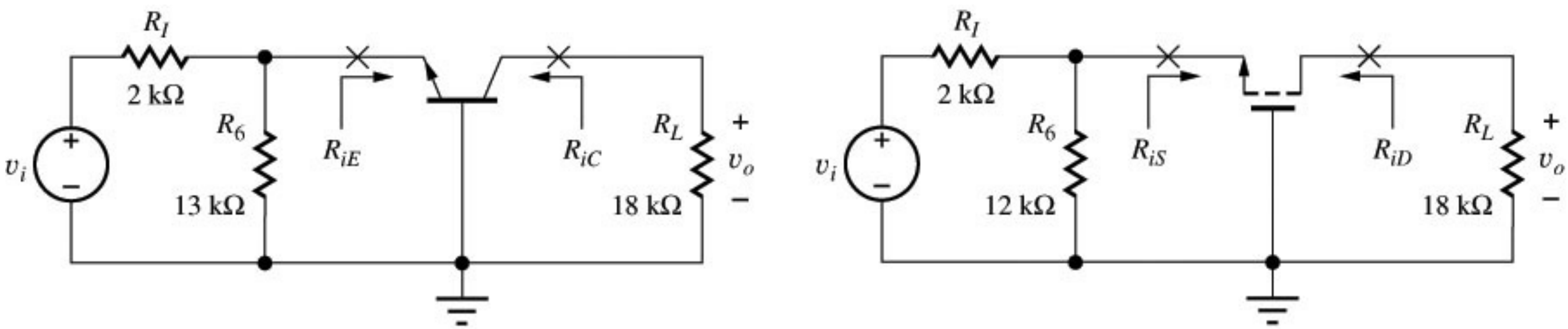
6.1.4 Common-Base/Gate Amplifier



Jaeger, Blalock, Fig. 14.5

- Signal injection into emitter/source, signal extraction from collector/drain
- Base/gate as common terminal between input and output ports; base/gate is connected to signal GND by bypass capacitors C_2

Common-Base/Gate Amplifier
– Small-Signal Equivalent Circuit –

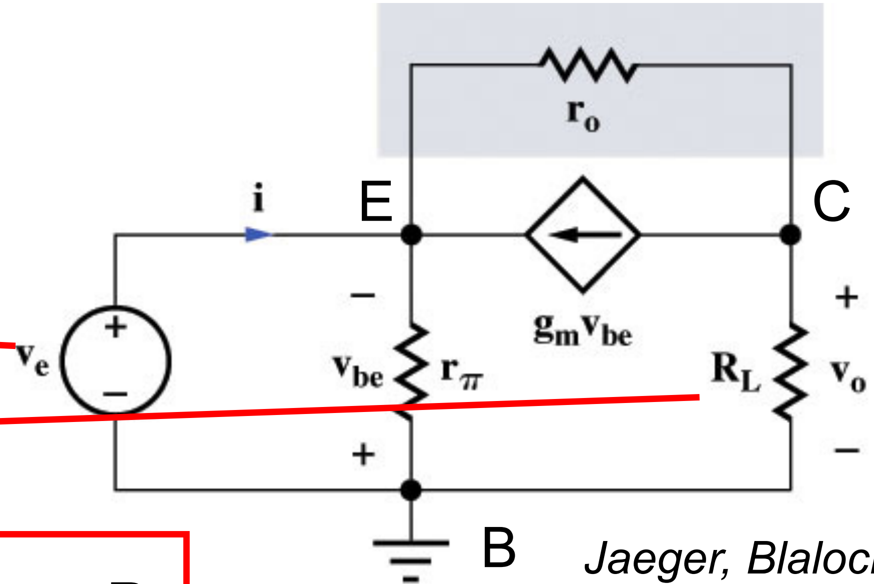


Jaeger, Blalock, Fig. 14.21

- BJT: $R_L = R_3||R_C = 18\ \text{k}\Omega$
  MOSFET: $R_L = R_3||R_D = 18\ \text{k}\Omega$
- C-B and C-G amplifiers provide **voltage gain and output resistance similar to that of C-E/C-S stages** but a **much lower input resistance**

Common-Base/Gate Amplifier
– Terminal Voltage Gain & Input Resistance –

- Analyze small equivalent circuit, neglecting the output resistance r_o, to find A_{vt}



Jaeger, Blalock, Fig. 14.22

$$v_e = -v_{be}$$

$$v_o = -g_m v_{be} R_L$$

$$\boxed{A_{vt}^{CB} = \frac{v_o}{v_e} = +g_m R_L \qquad A_{vt}^{CG} = +g_m R_L}$$

- Input resistance $R_{iE} = v_e/i$ requires knowledge of current i

$$i = -\frac{v_{be}}{r_\pi} - g_m v_{be} \quad \Rightarrow \quad \boxed{R_{iE} = \frac{v_e}{i} = \frac{1}{r_\pi^{-1} + g_m} \overset{g_m = \beta_0/r_\pi}{=} \frac{r_\pi}{\beta_0 + 1} \cong \frac{1}{g_m} \qquad R_{iS} = \frac{1}{g_m}}$$

Common-Base/Gate Amplifier
– Signal Source Voltage Gain –

- Calculated analogue to C-E voltage gain……

$$A_v^{CB} = \frac{v_o}{v_i} = \frac{v_o}{v_b}\frac{v_b}{v_i} = A_{vt}^{CB}\frac{v_b}{v_i} = A_{vt}^{CB}\frac{R_6 \| R_{iE}}{R_I + R_6 \| R_{iE}}$$

$$= \frac{g_m R_L}{1+g_m(R_6 \| R_I)}\frac{R_6}{R_I + R_6}$$

$$A_v^{CG} = \frac{g_m R_L}{1+g_m(R_6 \| R_I)}\frac{R_6}{R_I + R_6}$$

- Assuming $R_6 \gg R_I$, we obtain

$$\boxed{A_v^{CB} = A_v^{CG} = \frac{g_m R_L}{1+g_m R_I}}$$

Common-Base/Gate Amplifier
– Voltage Gain Limits –

$$\boxed{A_v^{CB} = A_v^{CG} = \frac{g_m R_L}{1 + g_m R_I}}$$

What are the limits for A_v?

(a) $g_m R_I \ll 1$: $A_v^{CB} \cong +g_m R_L$ $A_v^{CG} \cong +g_m R_L$

(b) $g_m R_{th} \gg 1$, $R_{th} = R_I || R_6$: $A_v^{CB} \cong +\frac{R_L}{R_I}$ $A_v^{CG} \cong +\frac{R_L}{R_I}$

Common-Base/Gate Amplifier
– Output Resistance –

- **Output Resistance:** Circuit analysis (similar to C-E amplifier!) yields for the output resistance ($R_{th} = R_I \,||\, R_6$)

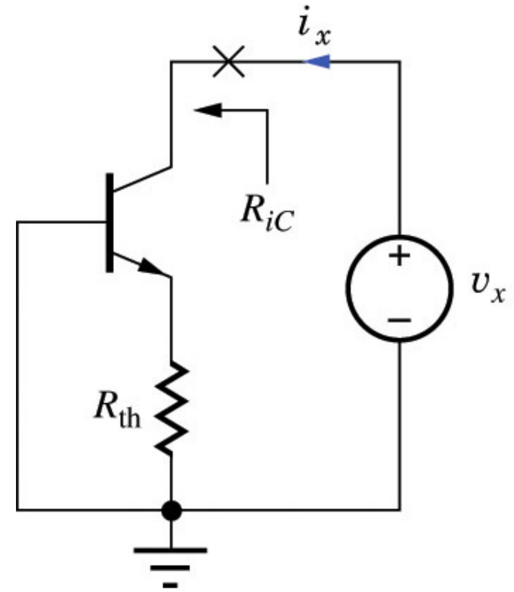
$$\boxed{R_{iC} = r_o\left(1 + \frac{\beta_0 R_{th}}{r_\pi + R_{th}}\right)}$$

- Using $\beta_0 = g_m r_\pi$, we find

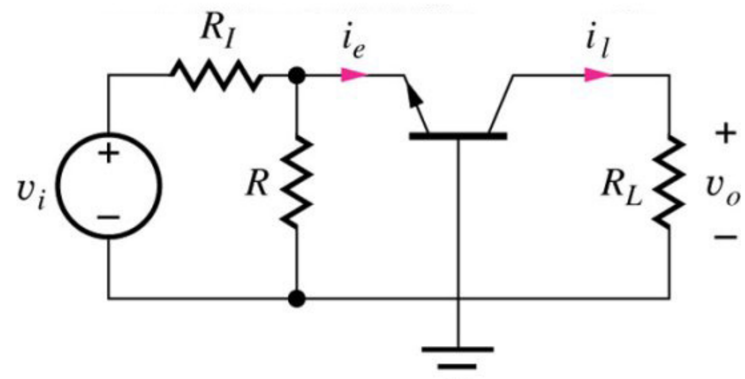
$$\boxed{R_{iC} = R_{iD} \cong r_o\left(1 + g_m(r_\pi \,||\, R_{th})\right) \cong \mu_f(r_\pi \,||\, R_{th})}$$

- **Current Gain:** Emitter and collector current are related by $i_C = \alpha_0 i_E$, i.e.

$$\boxed{A_{it}^{CB} = \frac{i_L}{i_e} = +\alpha_0 \cong 1 \qquad A_{it}^{CG} = 1}$$



Jaeger, Blalock, Fig. 14.23&24



Common-Base/Gate Amplifier
– Performance Summary –

	C-B Amplifier	C-G Amplifier
Terminal Voltage Gain A_{vt}	$A_{vt}^{CB} = +g_m R_L$	$A_{vt}^{CG} = +g_m R_L$
Signal Source Voltage Gain A_v	$A_v^{CB} = \dfrac{g_m R_L}{1+g_m R_{th}} \dfrac{R_6}{R_I + R_6}$	$A_v^{CG} = \dfrac{g_m R_L}{1+g_m R_{th}} \dfrac{R_6}{R_I + R_6}$
Terminal Current Gain A_{it}	$A_{it} = +\alpha_0 \approx 1$	$A_{it} = 1$
Input Resistance	$R_{iE} \cong g_m^{-1}$	$R_{iS} \cong g_m^{-1}$
Output Resistance	$R_{iC} \cong r_o\left(1+g_m R_{th}\right) = r_o + \mu_f R_{th}$	$R_{iD} \cong r_o\left(1+g_m R_{th}\right) = r_o + \mu_f R_{th}$
Input Signal Range	$\left\lvert v_b \right\rvert \le 0.005\left(1+g_m R_{th}\right)$	$\left\lvert v_g \right\rvert \le 0.2\left(V_{GS} - V_T\right)\left(1+g_m R_{th}\right)$

Common-Base/Gate Amplifier
– Performance Summary –

- C-B and C-G amplifiers feature significant voltage gain, low input resistance and high output resistance

- Values for the amplifier circuits on page 26:

	C-B Amplifier	**C-G Amplifier**
$g_m R_{th}$	16.8	0.84
Voltage Gain	+ 8.84	+ 4.11
Input Resistance	102 Ω	2.04 kΩ
Output Resistance	3.40 MΩ	411 kΩ
Input Signal Range	104 mV	422 mV
Current Gain	1	1

6.1.5 Single-BJT and Single-FET Amplifier Comparison

Single-BJT Amplifiers:

- Common-Emitter Amplifier: **inverting** amplifier, moderate to high voltage gain, moderate input resistance, output resistance and current gain; emitter resistor adds design flexibility (increase R_{in}, R_{out} and signal range but lose voltage gain)
- Common-Collector Amplifier: low voltage gain, high input resistance, low output resistance and moderate current gain
- Common-Base Amplifier: moderate to high voltage gain, low input resistance, high output resistance and low current gain

Single-FET Amplifiers:

- Common-Source Amplifier: **inverting** amplifier, moderate voltage gain and output resistance, high input resistance and current gain
- Common-Drain Amplifier: low voltage gain, high input resistance, low output resistance and high current gain
- Common-Gate Amplifier: moderate voltage gain, low input resistance, high output resistance and low current gain

Simplified Characteristics of Single-BJT Amplifiers

	C-E ($R_E = 0$)	C-E $R_E \neq 0$	C-C	C-B
Terminal Voltage Gain	$-\ g_m\ R_L$ $\approx -\ 10\ V_{CC}$ (high)	$-\ (R_L/R_E)$ (moderate)	1 (low)	$+g_m\ R_L$ $\approx +\ 10\ V_{CC}$ (high)
Input Resistance	r_π (moderate)	$\beta_0\ R_E$ (high)	$\beta_0\ R_L$ (high)	$1/g_m$ (low)
Output Resistance	r_o (moderate)	$\mu_f\ R_E$ (high)	$1/g_m$ (low)	$\mu_f\ (R_I\|\|R_6)$ (high)
Current Gain	$-\ \beta_0$ (moderate)	$-\ \beta_0$ (moderate)	$\beta_0 + 1$ (moderate)	1 (low)

For more details, consult Jaeger, Blalock, Table 14.9

Simplified Characteristics of Single-FET Amplifiers

	C-S ($R_E = 0$)	C-S $R_E \neq 0$	C-D	C-G
Terminal Voltage Gain	$-g_m R_L$ $\approx -V_{DD}$ (moderate)	$-(R_L/R_S)$ (moderate)	1 (low)	$+g_m R_L$ $\approx +V_{DD}$ (moderate)
Input Resistance	∞ (high)	∞ (high)	∞ (high)	$1/g_m$ (low)
Output Resistance	r_o (moderate)	$\mu_f R_S$ (high)	$1/g_m$ (low)	$\mu_f (R_I \| R_6)$ (high)
Current Gain	∞ (high)	∞ (high)	∞ (high)	1 (low)

For more details, consult Jaeger, Blalock, Table 14.11

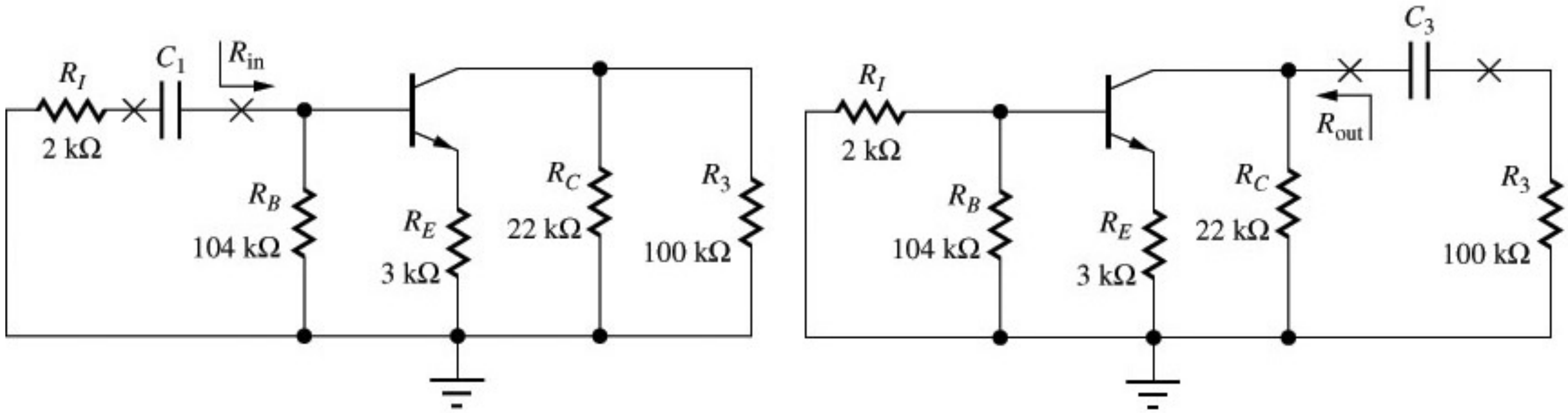
6.1.6 Coupling & Bypass Capacitors

- Finite impedance of coupling and bypass capacitors **limits amplifier gain at low frequencies**
- What capacitor values do we have to choose to ensure that the frequency response is "flat" down to a desired lower cut-off frequency?
- Example: Common-emitter amplifier with coupling capacitors C_1 and C_3 and by-pass capacitor C_2

Approach:

- Consider each capacitor separately and replace all other capacitors by short circuits ($C = \infty$)
- **To be able to neglect the capacitance "under test", the magnitude of its impedance must be much smaller than the equivalent resistance that appears at its terminals**

Common-Emitter Amplifier
– Coupling Capacitors C_1 and C_3 Design –



Jaeger, Blalock, Fig. 14.34

- Coupling Capacitor C_1: Impedance looking to left from C_1 is R_I, impedance looking to right is R_{in} (Note: R_{in} is not exactly the R_{iB} we calculated earlier)
$$\frac{1}{\omega C_1} \ll (R_I + R_{in}) \quad \Rightarrow \quad C_1 \gg \frac{1}{\omega(R_I + R_{in})}$$
- Coupling Capacitor C_3: Impedance looking to right from C_3 is R_3, impedance looking to left is R_{out} (Note: R_{out} is not exactly the R_{iC} we calculated earlier)
$$\frac{1}{\omega C_3} \ll (R_3 + R_{out}) \quad \Rightarrow \quad C_3 \gg \frac{1}{\omega(R_3 + R_{out})}$$

Common-Emitter Amplifier
– Coupling Capacitors C_1 and C_3 Design –

- Coupling Capacitor C_1: Bias resistor R_B appears in parallel with input resistance of transistor

$$\boxed{C_1 \gg \frac{1}{\omega(R_I + R_{in})} = \frac{1}{\omega(R_I + R_B \| R_{iB})}}$$

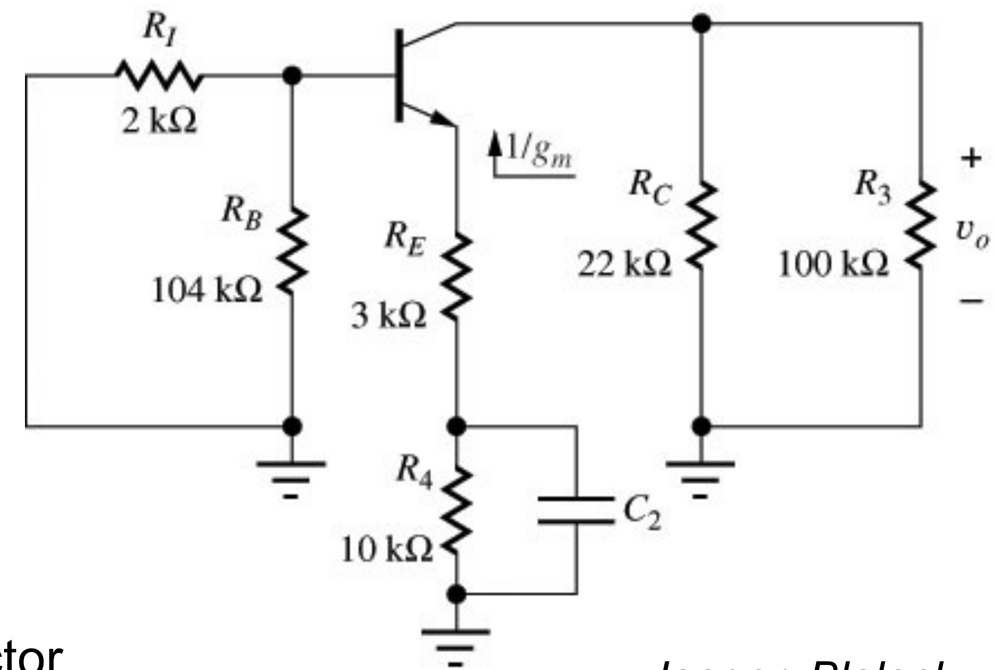
- Requiring that the capacitor C_1 can be neglected at e.g. 1 kHz, we find C_1 » 1.99 nF, i.e. we choose C_1 = 0.02 μF = 20 nF
- Coupling Capacitor C_3: Collector resistor R_C appears in parallel with output resistance of transistor

$$\boxed{C_3 \gg \frac{1}{\omega(R_3 + R_{out})} = \frac{1}{\omega(R_3 + R_C \| R_{iC})}}$$

- Requiring that the capacitor C_3 can be neglected at e.g. 1 kHz, we find C_3 » 1.31 nF, i.e. we choose C_3 = 0.015 μF = 15 nF

Common-Emitter Amplifier
– Bypass Capacitor C_2 Design –

- Neglect C_1 and C_3
- What is equivalent resistance at terminals C_2?
  R_4 in parallel with resistance looking upward toward emitter of transistors
- Calculating the resistance looking into the emitter similar to output resistance of common-collector amplifier ($R_{out} = g_m^{-1}$), we find

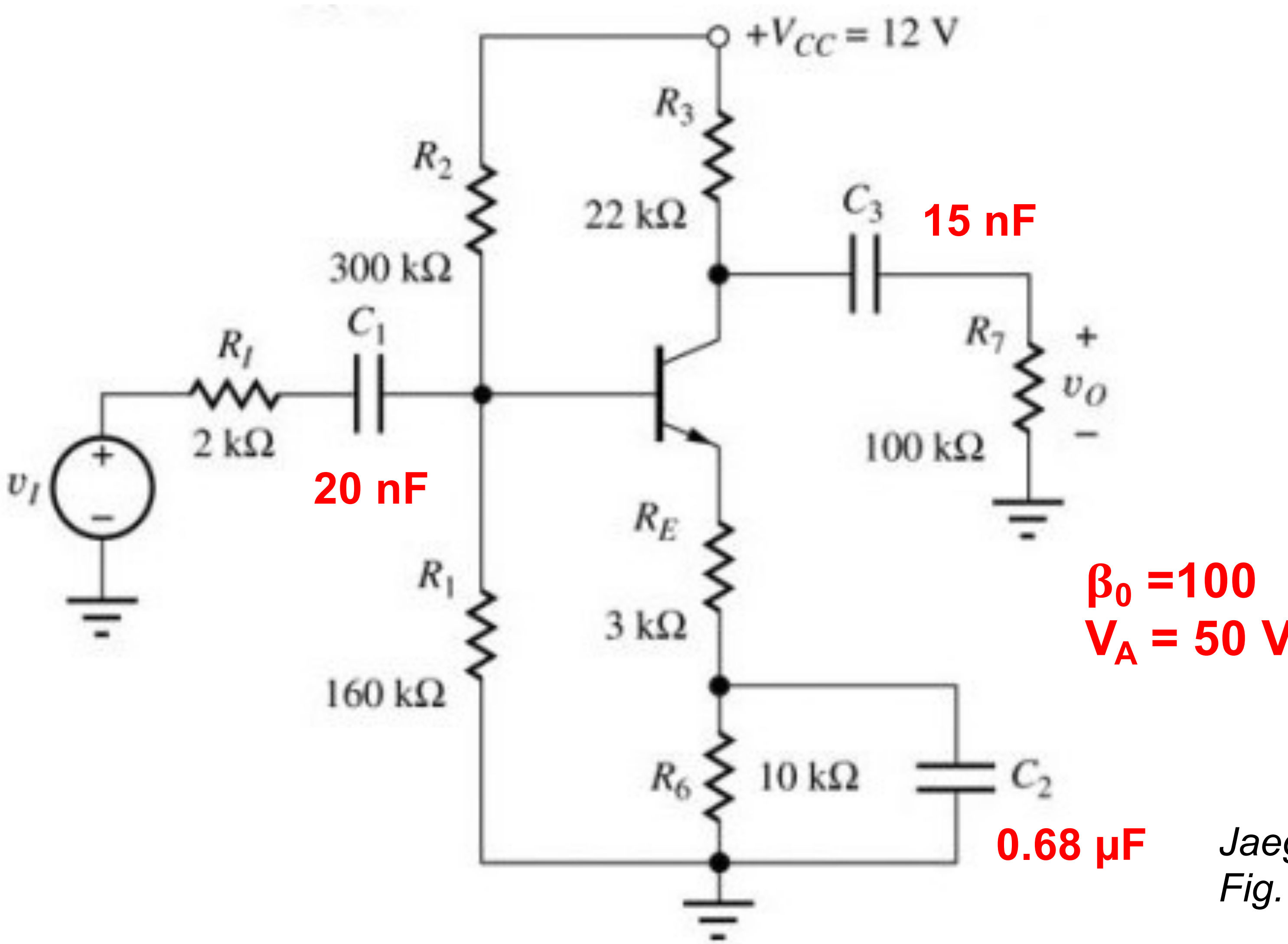


Jaeger, Blalock, Fig. 14.35

$$\boxed{\frac{1}{\omega C_2} \ll \left[R_4 \,||\, (R_E + \frac{1}{g_m}\right] \quad \Rightarrow \quad C_2 \gg \left\{\omega \left[R_4 \,||\, (R_E + \frac{1}{g_m}\right]\right\}^{-1}}$$

- Requiring that the capacitor C_2 can be neglected at e.g. 1 kHz, we find C_2 » 67.2 nF, i.e. we choose C_2 = 0.68 μF = 680 nF

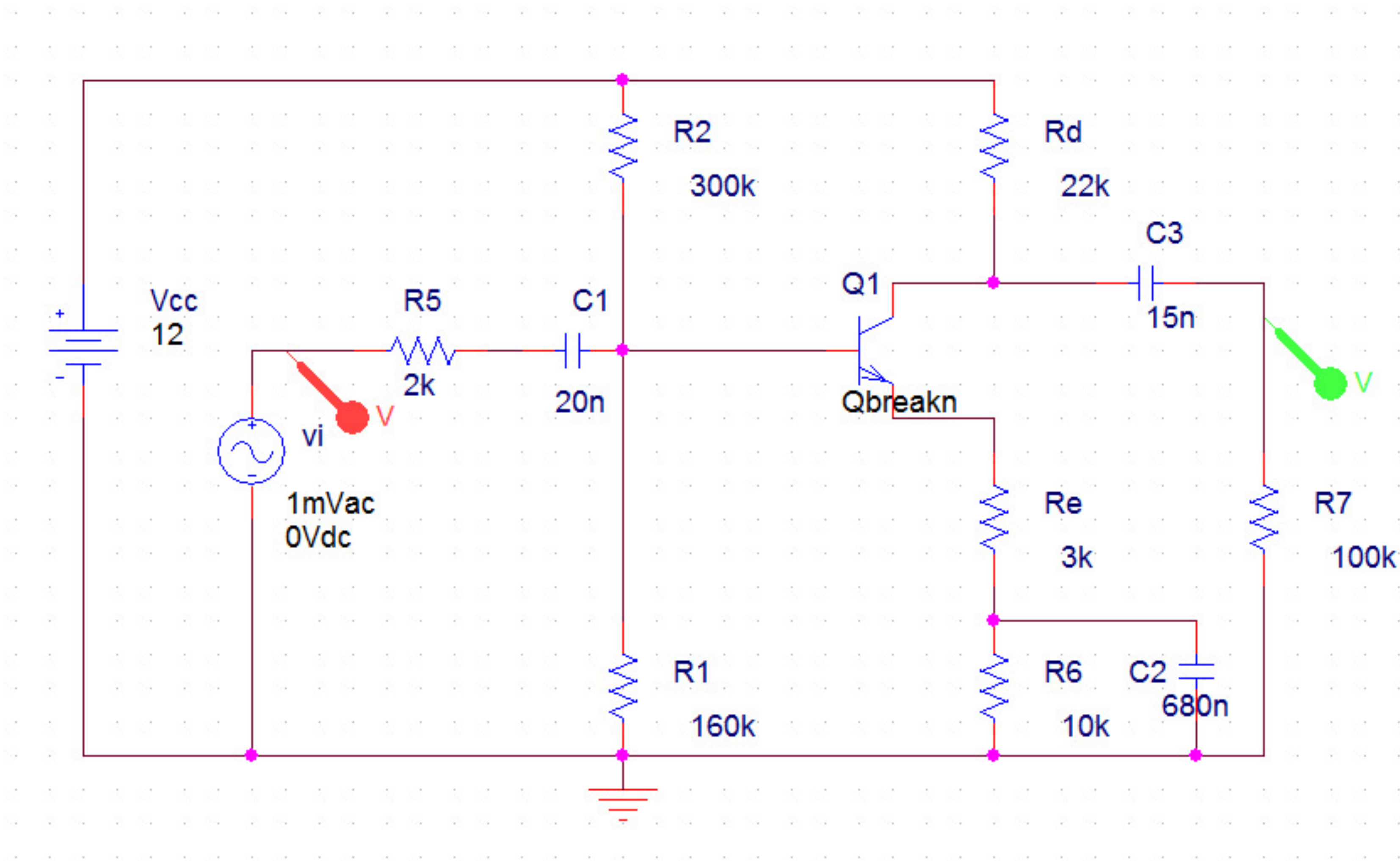
Common-Emitter Amplifier
– SPICE Simulation –



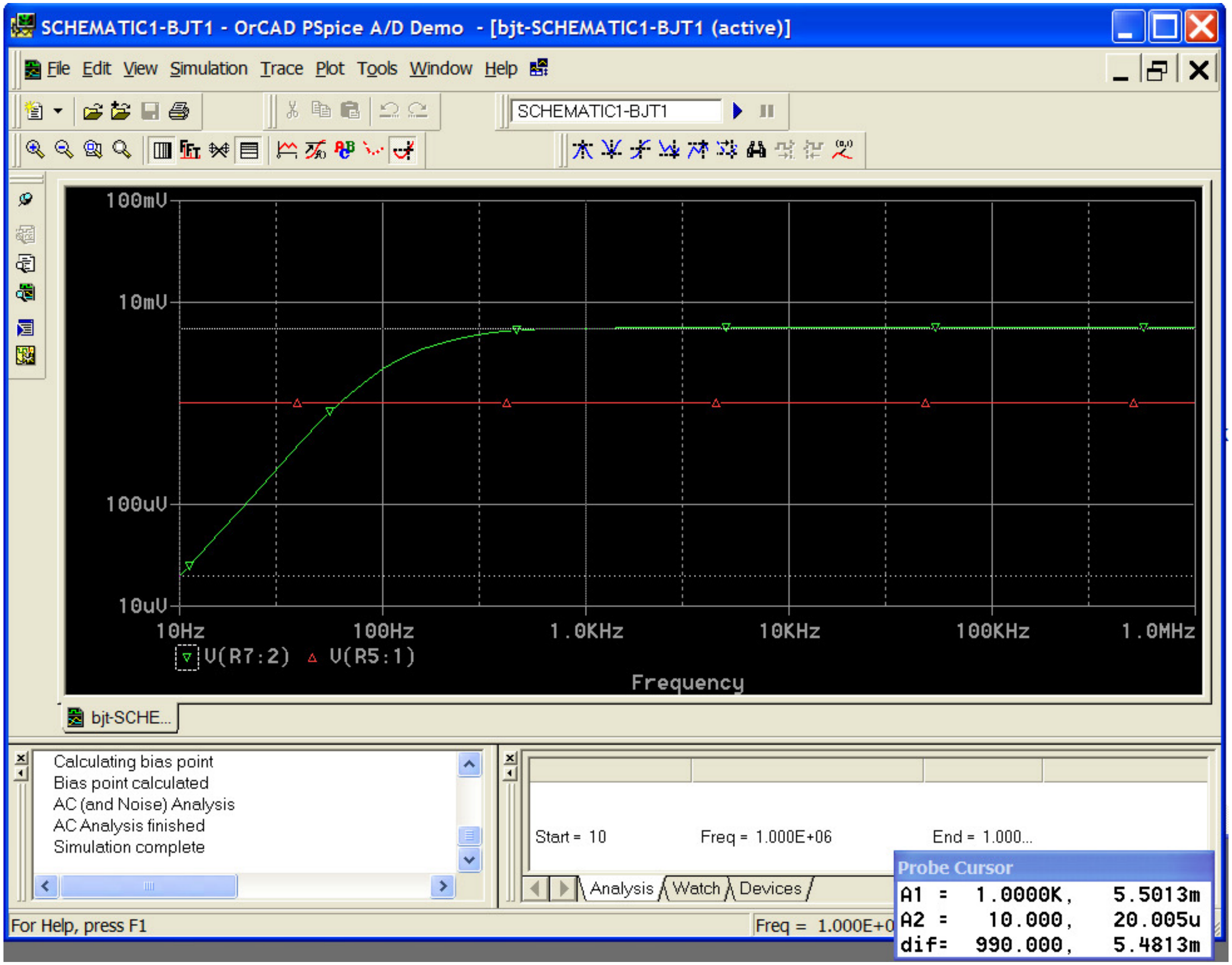
$\beta_0 = 100$
$V_A = 50$ V

Jaeger, Blalock, Fig. 14.2

Common-Emitter Amplifier
– SPICE Network –



Common-Emitter Amplifier
– SPICE Frequency Response –



Capacitor Design
– Single-Transistor Amplifiers –

	C_1	C_2	C_3
Common-Emitter	$C_1 \gg \frac{1}{\omega(R_I + R_B \parallel R_{iB})}$	$C_2 \gg \frac{1}{\omega\left[R_4 \parallel (R_E + \frac{1}{g_m}\right]}$	$C_3 \gg \frac{1}{\omega(R_3 + R_C \parallel R_{iC})}$
Common-Collector	$C_1 \gg \frac{1}{\omega(R_I + R_B \parallel R_{iB})}$		$C_3 \gg \frac{1}{\omega(R_3 + R_6 \parallel R_{iE})}$
Common-Base	$C_1 \gg \frac{1}{\omega(R_I + R_6 \parallel R_{iE})}$	$C_2 \gg \frac{1}{\omega\left[R_1 \parallel R_2 \parallel \left[r_\pi + (\beta_0 + 1)(R_6 \parallel R_I)\right]\right]}$	$C_3 \gg \frac{1}{\omega(R_3 + R_C \parallel R_{iC})}$